

Finding Hidden Functional States from EEG Based on Topological Features. Supplementary materials A.

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1. Supplementary materials. Hyperparameter values.

Table 1 (a) – Hyperparameter values for all methods used in the study.

Stage	Step	Description	Values
Feature engineering – processing each variable independently	Taken's embedding	Embedding dimension	[1; 20]
		The time delay between the coordinates of adjacent points	[1; 20]
		Stride	3
	Persistence diagram	Considered homology dimensions	{ 1, 2 }
Feature engineering – processing the correlations between the components	Sliding window	Size of the windows	50
		The stride of adjacent windows	20
	Persistence diagram	Considered homology dimensions	[1; 5]
Feature engineering – processing the EEG overall	Taken's embedding	Embedding dimension	5
		The time delay between the coordinates of adjacent points	11
		Stride	3
	Persistence diagram	Considered homology dimensions	{ 1, 2, 3 }
Extracting features from pers. diagrams	Filtering	The fraction of simplexes removed by filtering	10%
Feature selection	QSDA: stage 1	Number of clusters	[2, 15]
		Maximum distance between epochs in the connectivity matrix	[20, 50] with step 5
		Minimum stage length	{40}
		Coefficient of the Ward distance during stage merging	0.3

Table 1 (b) – Hyperparameter values for all methods used in the study.

Feature selection	QSDA: stage 2	The maximum number of clusters to construct the joint sets	{ 20 }
		The maximum distance between epochs to construct the joint sets	{ 50 }
		Number of clusters	[2, 10]
		Minimum stage length	{ 40 }
		Coefficient of the Ward distance during stage merging	0.2
Feature selection		Score threshold	At least 1500 features
		The minimum number of unique values required for the feature to be considered	200
Dimensionality reduction		The dimensionality after the reduction	15
SDA	Stage 1	Number of clusters	[2, 20]
		Maximum distance between epochs in the connectivity matrix	[20, 50]
		Minimum stage length	{ 0, 20, 40, 60 }
		Coefficient of the Ward distance during stage merging	0.3
SDA	Stage 2	The maximum value of $n_clusters$ to construct the joint sets	{ 10, 15, 20 }
		The maximum value of $k_neighbours$ to construct the joint sets	{35, 40, 45, 50}
		Number of clusters	[2, 15]
		Minimum stage length	{ 40 }
		Coefficient of the Ward distance during stage merging	0.2
Information value analysis		Number of containers when working with quantitative features	10